

External cognition

- Concerned with explaining how we interact with external representations (e.g. maps, notes, diagrams)
- What are the cognitive benefits and what processes involved
- How they extend our cognition
- What computer-based representations can we develop to help even more?

Externalizing to reduce memory load

- Diaries, reminders, calendars, notes, shopping lists, to-do lists
 - written to remind us of what to do
- Post-its, piles, marked emails
 - where placed indicates priority of what to do
- External representations:
 - Remind us that we need to do something (e.g. to buy something for mother's day)
 - Remind us of what to do (e.g. buy a card)
 - Remind us when to do something (e.g. send a card by a certain date)

Computational offloading

- When a tool is used in conjunction with an external representation to carry out a computation (e.g. pen and paper)
- Try doing the two sums below (a) in your head, (b) on a piece of paper and c) with a calculator.
 - $234 \times 456 = ??$
 - $CCXXXIII \times CCCCXXXXXVI = ???$
- Which is easiest and why? Both are identical sums

Annotation and cognitive tracing

- Annotation involves modifying existing representations through making marks
 - e.g. crossing off, ticking, underlining
- Cognitive tracing involves externally manipulating items into different orders or structures
 - e.g. playing Scrabble, playing cards

Design implication

- Provide external representations at the interface that reduce memory load and facilitate computational offloading
 - e.g. Information visualizations have been designed to allow people to make sense and rapid decisions about masses of data